# System Requirements and Parameters:

| **Input Group** | **Parameters Covered** |
|---|---|
| Environment | Wind speed, wind direction, temperature, RH |
| Machine | Travel speed, boom height |
| Spray system | Nozzle characteristics, spray pressure, flow rate/application rate |
| Spray behavior | Droplet sizes/class |

**System Assumptions:**

| Assumption | Project Definition |
|---|---|
| Target detection | A functioning weed/target-detection model is assumed to already exist. CoverageAI does not develop the weed-detection algorithm. |
| Boom width | Fixed at 120ft |
| Nozzle spacing | Fixed at 20in |
| Boom configuration | Physical boom architecture and nozzle spacing remain fixed during operation. |
| Boom height | Variable and measurable; may also be treated as a controllable parameter. |
| Nozzle options | Nozzle selection is limited to the five selected manufacturer application charts: TP65, ULD, 3D, XR, and DGX40. |
| Nozzle data source | Manufacturer application guides are assumed valid for nozzle-specific flow rate, droplet class, spray angle, recommended operating range, and related characteristics. |
| Nozzle lookup | Nozzle family + nozzle size + pressure are used to retrieve nozzle-dependent characteristics. |
| WSP feedback | One standardized WSP feedback card is placed at a known, repeatable location relative to the spray target. |
| WSP analysis | USDA-ARS DepositScan is used to obtain percent coverage and estimated deposition from WSP. |
| Pattern feedback | The WSP is treated as a **local spray-performance sensor**, not as a measurement of total field-scale drift. |
| Wind convention | Wind direction is assumed to be converted during |

| | |
|---|---|
| | sensing/preprocessing into the airflow-direction convention required by the model. |
| Sensor synchronization | Environmental, machine, and spray-system measurements are assumed to be time-synchronized with the spray event. |
| Operating limits | Inputs remain within the allowable ranges of the selected nozzle/application configuration. |
| Experimental consistency | WSP type, placement, orientation, scanning procedure, and DepositScan settings remain consistent across trials. |

**CoverageAI predicts and evaluates spray performance after the weed/target location is already known. Target identification itself is outside the scope of this model.**

## Parameter Classification

### Measured Inputs

| Variable | Unit | Data Type | Source |
|---|---|---|---|
| Wind speed | m/s | Continuous | Wind sensor |
| Wind direction | degree | Continuous/ circular | Wind direction sensor |
| Air temperature | Celsius | Continuous | Environmental sensor |
| Relative humidity | % | Continuous | Environmental sensor |
| Travel speed | km/h | Continuous | CAN/GPS |
| Boom height | m | Continuous | Boom height sensor |
| Spray pressure | PSI | Continuous | Pressure sensor |

### Configuration Inputs

| Variable | Unit / type | Data Type | Source |
|---|---|---|---|
| Nozzle family | Category | Categorical | Operator/ system configuration |
| Nozzle size | Manufacturer size code | Categorical | Operator/ system configuration |

Lookup / Derived inputs

| Variable | Unit | Type | Derived from |
|---|---|---|---|
| Flow rate | GPM or L/min | Continuous | Nozzle + size + pressure / flow sensor |
| Application rate | GPA or L/ha | Continuous | Flow + travel speed + treated width |
| Droplet class | ASABE spray class | Ordinal/ categorical | Nozzle + size + pressure |
| Spray angle | degree | Continuous / configuration | Nozzle guide |
| Nozzle technology | category | categorical | Nozzle guide |
| Relative wind angle | degree | Continuous | Wind direction + vehicle heading |
| Crosswind component | m/s | Continuous | Wind speed + relative wind angle |
| Along-track wind | m/s | Continuous | Wind speed + relative wind angle |

**Nozzle Lookup Architecture**

$$Nozzle\ family\ +\ nozzle\ size\ +\ pressure$$
$$\downarrow$$
$$Manufacturer\ lookup\ table$$
$$\downarrow$$
$$flow\ rate\ +\ droplet\ class\ +\ spray\ angle\ +\ nozzle\ technology$$

**CoverageAI Model Inputs**

$$X\ =\ [U,\ \theta_{rel},\ U_{cross},\ T,\ RH,\ v,\ H,\ P,\ Q,\ AR,\ droplet\ class,\ \alpha,\ nozzle\ technology$$

where:

- U = wind speed
- $\theta_{rel}$= relative wind direction
- Ucross = crosswind component
- T = temperature
- RH = relative humidity
- v = travel speed
- H = boom height
- P = spray pressure
- Q = flow rate
- AR = application rate
- a = spray angle

**Validated outputs**
CoverageAI will predict three primary spray-performance outcomes

| Output | Unit | Validation |
|---|---|---|
| Spray coverage | % area covered | WSP + USDA DepositScan |
| Estimated target deposition | $uL/cm^2$ or DepositScan equivalent | WSP + USDA DepositScan |
| Spray pattern deviation / displacement | mm/cm and/or coverage deviation | Expected pattern vs. observed WSP pattern + environmental context |

Conceptually:

$$Y = [C, D, \Delta r]$$

where:

- C = coverage
- D = estimated deposition
- $\Delta r$ = local pattern displacement/ deviation

**Prediction and feedback structure**

$$environmental\ sensors + machine\ state + spray\ system\ state + nozzle\ lookup$$
$$\downarrow$$
$$CoverageAI\ Feature\ Vector$$
$$\downarrow$$
$$Predict$$
$$\{\widehat{C}, \widehat{D}, \widehat{\Delta r}$$
$$\downarrow$$
$$Spray$$
$$\downarrow$$
$$WSP + DepositScan$$
$$\{C_{obs}, D_{obs}, \Delta r_{obs}$$
$$\downarrow$$
$$Compare\ prediction\ vs\ observation$$

**Model Validation**

Each predicted output can be compared against the post spray measurement

For coverage: $e_C = C_{obs} - \widehat{C}$

For deposition: $e_D = D_{obs} - \widehat{D}$

For displacement $e_r = \Delta r_{obs} - \widehat{\Delta r}$

Absolute error calculated as $AE = |Y_{obs} - \widehat{Y}|$

**Controllable vs Non controllable**

| Parameters | System can change it |
|---|---|
| Wind speed | no |
| Wind direction | no |
| temperature | no |
| Relative humidity | no |
| Travel speed | yes |
| Boom height | Yes, assuming active height control |
| Nozzle family | Yes between configurations, generally not during a pass |
| Nozzle size | Yes between configurations, generally not during the pass |
| Spray pressure | yes |
| Flow rate | yes/ indirectly controlled |
| Application rate | yes/ derived from controllable variables |
| Droplet characteristics | Indirectly through nozzle and pressure |

**Closed-loop control concept**

Eventually predicted poor performance could trigger an adjustment of $v,\ H,\ P,\ Q$

So:

$$sense\ \rightarrow\ predict\ \rightarrow optimize \rightarrow spray \rightarrow verify \rightarrow update$$

# Detailed Parameter Definitions

## Wind Speed:

| Field | Definition |
|---|---|
| Parameter | Wind Speed |
| Category | Environment |
| Unit | m / s |
| Role | Continuous model input / predictive feature |
| Controllable? | No |
| Measured or Derived? | Measured |
| Expected Range | Proposed modeling range: 0-7 m/s (~0-15.7 mph). Actual allowable application range is label dependent. EPA guidance commonly used wind-speed limits in the 10-15 mph range depending on the application and label. |
| Primary Relationships | Increasing wind speed generally increases horizontal droplet displacement and drift potential. Its effect depends strongly on droplet size/ settling velocity, release height, and wind direction. |
| Output Affected | Primarily off-target deposition/drift; secondarily target deposition, coverage, and coverage uniformity. |
| Source/ Data Needed | On-machine or field anemometer; synchronized wind measurement during spraying. Wind should ideally be measured at or above spray release height and away from major obstructions. |

**Measurement Definition:**

Wind speed should represent ambient air velocity near the spray release height during application, rather than distant weather-station measurement. EPA guidance states that wind speed used for spray-drift decisions should be measured at release height or higher and in an area free from obstructions such as trees, buildings, and equipment.

**Simplified Physical Relationship**

The first-order relationship we decided to use is:

$$x \cong U\ (H/Vt)$$

=

where:

- x = estimated horizontal droplet displacement, m
- U = horizontal wind speed, m/s
- H = droplet release height above the target, m
- Vt = terminal settling velocity of the droplet, m/s

This comes from combing:

$$x = Ut$$

with:

$$t = H / Vt$$

therefore:

$$x = U\,(H / Vt)$$

**Interpretation:**

This relationship tells us:

$$U \uparrow \Rightarrow x \uparrow$$

Higher wind speed increases horizontal displacement

$$H \uparrow \Rightarrow x \uparrow$$

Greater release height gives the droplet more time to move horizontally.

$$Vt \uparrow \Rightarrow x \downarrow$$

Droplets with greater settling velocity reach the target faster and therefore have less time to drift.

**Droplet settling velocity reference**
Rather than calculating terminal velocity ourselves, let's use published droplet settling velocities.

**Table II.7–1—Selected characteristics of various size spray droplets of water**

| Droplet diameter | Terminal velocity | Fall time from 3 m | Drift distance (3-m fall with 5-km/h wind) | Drops/cm² from 10 a/ha application |
|---|---|---|---|---|
| *(μm)* | *(M/sec)* | *(Sec)* | *(M)* | *(No./cm²)* |
| 10 | 0.003 | 1,020 | 1,372 | 190,990 |
| 50 | 0.075 | 40 | 54 | 1,530 |
| 100 | 0.279 | 11 | 15 | 192 |
| 200 | 0.721 | 5.4 | 5 | 24 |
| 500 | 2.139 | 1.6 | 2 | 1.5 |

The same USDA table also demonstrates the relationship between droplet diameter, fall time, wind speed, and drift distance, making it a useful supporting source for the simplified displacement approach.

**Assumptions of this relationship**
The equation

$$x \cong U\,(H/Vt)$$

should be described as a first-order or simplified drift-distance approximation, not as the project's final drift model.

It assumes:

- Constant wind speed
- Constant droplet settling velocity

- Approximately horizontal wind
- No evaporation
- No turbulence
- No changing droplet diameter
- No canopy interception
- No wake effects from the sprayer
- No detailed nozzle exit velocity effects

So the purpose of this equation is to show the expected physical relationship between variables, not to perfectly predict real-world drift.

**Conclusion**: Wind speed is a continuous environmental input measured in m/s near spray release height. Increasing wind speed generally increases off-target droplet displacement, with the magnitude of the effect strongly influenced by release height and droplet settling velocity. A first-order displacement relationship is $x \cong UH / Vt$ , which may be used as a physics-informed feature or conceptual baseline rather than as the final drift model.

Sources:
https://www.epa.gov/pesticides/mitigation-menu-measure-descriptions
https://www.ars.usda.gov/ARSUserFiles/30320505/grasshopper/Extras/PDFs/IPM%20Handbook/SectionII-Chemical.pdf
https://www.epa.gov/pesticide-registration/prn-2001-x-draft-spray-and-dust-drift-label-statements-pesticide-products

# Wind Direction:

| Field | Definition |
|---|---|
| Parameter | Wind Direction |
| Category | Environment |
| Unit | Degrees (°), preferably converted to a relative angle or vector components for modeling |
| Role | Continuous environmental input / directional predictive feature |
| Controllable? | No |
| Measured or Derived? | Measured, then optionally derived into relative-angle or vector components. |
| Expected Range | 0° - 360° |
| Primary Relationships | Determines the direction in which spray droplets are transported; interacts directly with wind speed, sprayer travel direction, field orientation, and location of off-target/ sensitive areas |
| Output Affected | Primarily direction and spatial distribution of off-target deposition/ drift; also target deposition and coverage distribution |
| Source/ Data Needed | Wind-direction sensor plus vehicle heading/GPS |

**Measurement Definition:**
For this project, we will assume the sensor/preprocessing layer provides the direction the airflow is moving toward, so the model does not need a separate meteorological “from-direction” conversion

$$\theta w \;=\; airflow\ direction$$

with:

- 90° = wind toward east
- 180°= wind toward south
- 270° = wind toward west
- 360° = wind toward north

**Relative Wind Direction:**
Absolute compass direction is less useful to the model than the direction of airflow relative to the machine’s travel direction.

Let:

$$\theta w \;=\; wind\ angle$$

and

$$\theta v \;=\; vehicle\ heading$$

Then we define the relative angle as:

$$\theta rel = \theta w - \theta v$$

The results can then be normalized to a range such as:

$$-180° \leq \theta rel \leq 180°$$

EPA spray-drift field guidance supports the importance of wind direction relative to machine travel. EPA specifies that deposition collectors should be arranged appropriately parallel to the wind direction and as close as possible to perpendicular to the direction of application equipment travel.

Wind-vector Components:
Once wind speed U and relative direction $\theta rel$ are known, the wind can be separated into:

- A component parallel to machine travel
- A component perpendicular to machine travel

If U is wind speed and $\theta rel$ is the angle relative to the vehicle path, then:

$$Ualong = Ucos(\theta rel)$$

and

$$Ucross = Usin(\theta rel)$$

Formulas derived from standard vector-component formulas with the vehicle travel direction chosen as one axis.

**Interpretation:**

If:

$$\theta rel = 0°$$

Then:

$$Ucross = 0$$

And all wind is parallel to travel.

If:

$$\theta rel = 90°$$

Then:

$$Ucross = U$$

And wind is entirely perpendicular to travel.

**Connection to the simplified drift relationship**

For wind speed, the first-order relationship is:

$$x \cong U\ (H/Vt)$$

For cross-track displacement, it is more meaningful to use only crosswind component:

$$x\ cross \cong Ucross(H/Vt)$$

Substituting:

$$Ucross = Usin(\theta rel)$$

Gives:

$$x\,cross \cong Usin(\theta rel)(H/Vt)$$

This is a project-derived first-order approximation, not EPA or published pesticide-drift equation. EPA's role here is supporting the physical relevance of wind orientation relative to machine travel and downwind deposition, not supplying this exact equation.

**Recommended model treatment**

For the raw dataset, retain:

Wind Direction
Wind Speed
Vehicle Heading

Then derive:

$\theta rel$
$Ucross$
$Ualong$

The derived components are likely more useful to the model than raw compass direction alone because they describe airflow relative to the spraying operation.

**Conclusion**: Wind direction is a continuous environmental input representing the direction in which ambient airflow is moving. For this project, any conversion from meteorological "from" direction is assumed to occur at the sensor or preprocessing stage. Wind direction is compared with vehicle heading to calculate relative wind angle. Standard vector decomposition is then used to determine crosswind and along-track wind components, which influence direction and spatial distribution of spray deposition and drift.

Sources:
https://forecast.weather.gov/glossary.php?word=DIR
Spray Drift Test Guidelines OPPTS 840.1200 Spray Drift Field Deposition

# Temperature & Relative Humidity:

| Field | Air Temperature | Relative Humidity |
|---|---|---|
| Category | Environment | Environment |
| Unit | °C | % |
| Role | Continuous model input | Continuous model input |
| Controllable? | No | No |
| Measured or Derived? | Measured | Measured |
| Expected Range | 10 - 40°C | 10 - 100 % RH |
| Primary Relationships | Higher temperature generally increases droplet evaporation, especially when relative humidity is low | Lower RH generally increases droplet evaporation, especially when temperature is high |
| Output Affected | Off-target deposition/drift, target deposition, coverage | Off-target deposition/drift, target deposition, coverage |
| Source/ Data Needed | Ambient temperature sensor | Relative-humidity sensor |

**Why they are grouped together:**

Temperature and relative humidity are most useful in this project because they jointly influence spray-droplet evaporation. University of Nebraska Extension states that low relative humidity and/or high temperature cause faster evaporation of droplets and increasing drift potential. It also notes that hot, dry conditions can reduce deposition on the target.

Key relationship for the model is:

$$T\uparrow,\ RH\downarrow \Rightarrow evaporation\ rate\uparrow$$

Followed by:

$$evaporation\uparrow \Rightarrow d\downarrow$$

Where d is droplet diameter.

**Connection to drift**

As a droplet evaporates, its diameter decreases. Smaller droplets are more easily transported by ambient air currents and generally remain suspended longer.

So our conceptual chain is:

$$T\uparrow,\ RH\downarrow \rightarrow evaporation\uparrow \rightarrow d\downarrow \rightarrow Vt\downarrow \rightarrow drift\ susceptibility\uparrow$$

This ties directly back to the wind-speed relationship we already established:

$$x \cong U\ (H/Vt)$$

If evaporation reduces settling velocity Vt, then the droplet remains airborne longer and has more time to move horizontally. This should be treated as a physical relationship and model rationale, not a complete evaporation model.

**Experimental Support:**
Zhu et al. simulated spray-droplet drift using:

$$10℃ \leqq T \leqq 30℃$$

And:

$$10\% \leqq RH \leqq 100\%$$

along with wind velocity, droplet size, release height, initial droplet velocity, and turbulence. They found that temperature and humidity had a much greater influence on the drift distance of droplets smaller than 100 um than on droplets 200 um and larger. Their simulations also found that, except under low-temperature/high humidity conditions, droplets around 50 um and smaller could completely evaporate before depositing over relatively short vertical distances.

This means the interaction is:

$$T\ x\ RH\ x\ d$$

**Droplet-size sensitivity**
University of Florida IFAS gives a useful practical breakdown:

- Droplets $\leqq$ 50 um may evaporate very rapidly before reaching the target
- Droplets $>$200 um generally experience much less size reduction before deposition
- Droplets roughly 50-200 um are particularly influenced by temperature, humidity, and other weather conditions

**Conclusion**: Air temperature and relative humidity are separate continuous environmental inputs documented together because of their combined influence on droplet evaporation. High temperature and low relative humidity increases evaporation, reducing droplet diameter and increasing susceptibility to off-target transport. Their effect is strongly dependent on droplet size, with smaller droplets being substantially more sensitive than large droplets. Temperature and relative humidity therefore primarily influence off-target deposition/ drift, target deposition, and coverage through their effect on droplet evaporation.

**Sources**:
https://extensionpubs.unl.edu/publication/g1773/na/html/view
https://elibrary.asabe.org/abstract.asp?aid=28220&

# Travel Speed:

| Field | Definition |
|---|---|
| Parameter | Travel Speed / Ground Speed |
| Category | Machine / Operational |
| Unit | Km/h or m/s |
| Role | Continuous model input and controllable operating parameter |
| Controllable? | Yes |
| Measured or Derived? | Measured |
| Expected Range | Use the actual operating range of the machine |
| Primary Relationships | Directly interacts with nozzle flow rate, spray width/nozzle spacing, and application rate. If flow is held constant, increasing travel speed decreases spray volume applied per unit area. |
| Output Affected | Application rate, target deposition, spray coverage, coverage uniformity; potentially off-target deposition indirectly |
| Source/ Data Needed | Machine CAN bus, wheel-speed sensor, GPS/GNSS, or other synchronized machine-speed measurement |

**ASABE basis:**

The primary standard for this parameter is ASAE EP367.2 – Guide for Preparing Field Sprayer Calibration Procedures. ASABE states that the Engineering Practice provides guidance for calibrating boom-type field sprayers used for broadcast, band, or row applications, with the goal of improving the uniformity, accuracy, and safety of pesticide application. Travel speed is a fundamental calibration variable because it determines how much ground area is covered during a given period of spray flow.

**Critical application-rate relationship**

For a broadcast boom sprayer in SI units:

$$q = (ARws) / 600$$

where:

- $q$ = total boom flow rate, L/min
- $AR$ = application rate, L/ha
- $w$ = treated boom width, m
- $s$ = travel speed, km/h

Rearranged to solve for application rate gives:

$$AR = (600q) / ws$$

**Primary travel-speed relationship:**
If boom width and flow remain constant:

$$AR = (600q) / ws$$

therefore:

$$AR \propto 1 / s$$

So:

$$s \uparrow \Rightarrow AR \downarrow$$

Only when q and w remain constant

**Rate-control interaction**
A modern closed-loop or rate-controlled sprayer can respond to increased speed by increasing flow:

$$s \uparrow \Rightarrow q \uparrow$$

Such that:

$$AR = constant$$

From the ASABE relationship:

$$q = (ARws) / 600$$

If AR and w are fixed:

$$q \propto s \text{ when } AR \text{ and } w \text{ are constant}$$

So if travel speed rises by 20%, the required total flow must also rise by approximately 20% to maintain the same application rate.

**Connection to deposition and coverage**
Relationship:

$$s \rightarrow AR \rightarrow deposition/coverage$$

So:

Without rate compensation

$$s \uparrow \Rightarrow AR \downarrow \Rightarrow less\ spray\ volume\ per\ unit\ area$$

With rate compensation

$$s \uparrow \Rightarrow q \uparrow \Rightarrow AR \cong constant$$

**Recommended Model Treatment**
Travel speed should remain a continuous variable:

$$s = ground\ speed\ in\ km/h$$

The primary interactions to retain are:

$$s\ x\ q$$

Travel speed and total flow rate,
and:

$$s\ x\ w$$

travel speed and width.
For the eventual closed-loop controller, travel speed could also be one of those variables the system adjusts in response to predicted coverage or deposition performance.

**Conclusion**: Travel speed is a continuous and controllable machine parameter representing the forward ground velocity of the sprayer. ASAE367.2 establishes field sprayer calibration as a process involving interacting application variables, and ASABE engineering references express application flow using $q = ARws/600$. At constant boom width and flow rate, application rate is inversely proportional to travel speed. To maintain a constant application rate as travel speed changes, flow rate must be adjusted accordingly. . Travel speed should therefore be modeled jointly with flow rate and spray width rather than treated as an independent predictor of deposition or coverage.

Sources:
https://elibrary.asabe.org/abstract.asp?aid=24435&redir=&redirType=&t=28

# Boom Height:

| Field | Definition |
|---|---|
| Parameter | Boom height / spray release height |
| Category | Machine / Spray system |
| Unit | m |
| Role | Continuous model input and potentially controllable operating parameter |
| Controllable? | Yes, if the machine has active boom-height control |
| Measured or Derived? | Measured |
| Expected Range | Equipment and nozzle dependent. 0.3-0.75 m above target/canopy based on published ASABE field testing. |
| Primary Relationships | Increasing boom height increases droplet airborne time and therefore potential wind displacement; boom height also affects spray-pattern overlap and coverage uniformity |
| Output Affected | Off-target deposition/drift, target deposition, coverage, coverage uniformity |
| Source/ Data Needed | Boom-height sensor, ultrasonic/radar sensor, machine CAN data, or known commanded boom height. |

ASABE field research by Nuyttens et al. evaluated boom heights of 0.30, 0.50, and 0.75 m and found that lower boom heights generally reduced spray drift. Their reference configuration used a 0.50m boom height and 0.50m nozzle spacing

**Why boom height matters**:
Boom height affects the amount of time spray droplets spend suspended between the nozzle and the target. The EPA applicator manual states directly that the higher the boom is above the target, the greater the potential for drift, and that lowering the boom can reduce off-target movement.

EPA also currently distinguishes between:

$$Low\ boom \leq 24\ in$$

and

$$High\ boom > 24\ in$$

Above ground or crop canopy for ground-boom drift mitigation. For fine-droplet applications, lowering the boom to 2 ft or less can qualify for substantial drift buffer reduction. However, for this model, we should retain continuous height.

**Main physical relationship**

If a droplet falls a vertical distance H at settling velocity Vt, then its approximate airbornet time is:

$$t = H / Vt$$

Where:

- t = approximate airborne time, s
- H = boom/release height above target, m
- Vt = droplet settling velocity, m/s

This means:

$$H \uparrow \Rightarrow t \uparrow$$

A higher boom gives the droplet more time in the air.
Then horizontal displacement due to wind is approximated by:

$$x = Ut$$

Substituting the fall-time relationship:

$$x \cong U\,(H / Vt)$$

where:

- x = horizontal droplet displacement, m
- U = wind speed, m/s

Therefore, if wind speed and droplet settling velocity stay constant

$$x \propto H$$

Simplified in this model:

$$H \uparrow \Rightarrow xdrift \uparrow$$

So increasing height by 50% produces approximately 50% more displacement under the assumptions of this simple model.

**Spray Pattern Overlap**
Boom height has another important effect besides drift: it changes how adjacent nozzle spray patterns overlap at the target. Flat-fan nozzles create expanding spray patterns as distance from the nozzle increases. Boom height, nozzle spacing, and spray angle therefore work together to determine overlap.

Conceptually:

$$Boom\ Height + Nozzle\ Angle + Nozzle\ Spacing \rightarrow Pattern\ Overlap$$

This matters because inadequate overlap can cause streaking, underapplication between patterns, and poor coverage uniformity. So there is a tradeoff, if boom is to low then potential insufficient overlap while boom to high can lead to greater drift potential.

The model should receive boom height, nozzle type/ spray angle, and nozzle spacing then learn their relationship to coverage.

**Optional spray geometry**
Here is a simple geometric relationship for the data book where a spray fan with included angle a at height H has an approximate theoretical width:

$$Ws = 2H\ tan\,(\alpha/2)$$

where:

- Ws = theoretical spray pattern width
- H = nozzle height above target
- a = spray angle

So:

$$H \uparrow \Rightarrow Ws \uparrow$$

For a fixed spray angle

**Boom height x nozzle spacing**
Important practical relationship is:

$$Pattern\ overlap\ depends\ on\ (Ws\ /\ S)$$

where:

- Ws = spray fan width at target
- S = nozzle spacing

**Interaction with wind and droplet size**
Boom height should not be modeled independently.

Its drift effect depends heavily on:

$$H\ x\ U$$

Boom height x wind speed
And:

$$H\ x\ Vt$$

Boom height x droplet settling velocity

The combined first-order relationship is still

$$x \cong (UH)\ /\ Vt$$

This tells the model that raising the boom is especially problematic when wind speed is high and droplets are small or have a low settling velocity.

**Recommended model treatment**
Store boom height as:

$$H\ =\ continuous\ height\ above\ target/canopy\ in\ meters$$

Conclusion: Boom height is a continuous and potentially controllable spray-system parameter representing the vertical release height of the nozzle above the target or crop canopy. Increasing boom height increases droplet airborne time, increasing the opportunity for wind-induced displacement and off-target deposition. ASABE field research evaluating boom heights of 0.30, 0.50, and 0.75 m found that lower boom heights generally reduced spray drift. Boom height also interacts with nozzle spray angle and spacing to determine spray-pattern overlap and coverage uniformity. The parameter should therefore be modeled jointly with windspeed, droplet size/ settling velocity, nozzle characteristics, and target height.

Sources:
https://elibrary.asabe.org/abstract.asp?aid=23622&utm
Pesticide Applicator Certification Core Manual Second Ediditon

# Nozzle Characteristics:

| Field | Definition |
|---|---|
| Parameter | Nozzle Characteristics |
| Category | Spray system / Configuration |
| Unit | Mixed: categorical, degrees, PSI, GPM, droplet class |
| Role | Configuration input and lookup source for derived spray characteristics |
| Controllable? | Partially – nozzle selection can be changed between applications, but generally not dynamically during a spray pass. |
| Measured or Derived? | Primarily configured and lookup-derived |
| Expected Range | Limited to the five selected nozzle families and manufacturer-supported size/operating ranges |
| Primary Relationships | Nozzle family, nozzle size, and pressure determine flow rate, droplet size classification, spray pattern, and ultimately influence coverage, deposition, and drift |
| Output Affected | Flow rate, droplet characteristics, application rate, coverage, target deposition, spray pattern deviation |
| Source/ Data Needed | Manufacturer nozzle/application guides and operating charts |

**Purpose**:
Nozzle characteristics describe the spray hardware configuration and provide the model with information about expected spray behavior. Rather than treating the nozzle model name itself as the main predictive feature, the system will use the selected nozzle as a lookup key to retrieve physically meaningful characteristics from manufacturer application charts.

The five nozzle families currently selected for the project are:

- TeeJet TP65
- Hypro ULD
- Hypro 3D
- TeeJet XR
- TeeJet DGX40

The exact nozzle size within each family will also be specified.

**Core system relationship:**

The main relationship we are using is:

$$Nozzle\ Family\ +\ Nozzle\ Size\ +\ Pressure\ \rightarrow\ Flow\ Rate\ +\ Droplet\ Class\ +\ Spray\ Characteristics$$

**Raw configuration inputs:**

These are the values the system would know directly.

1. Nozzle Family
    a. Example: TP65, ULD ,3D, XR, DGX40
    b. Data Type: categorical
    c. Purpose: identifies which manufacturer performance table should be used.
2. Nozzle Size
    a. Example: 02, 03, 04, 05, 06, 08…
    b. Nozzle size identifies the flow-capacity class of the nozzle
    c. Data type: categorical or ordinal configuration value
    d. Strongly affects flow rate, droplet class, and application rate at a given speed
3. Spray Pressure
    a. Is its own parameter but is required for the nozzle look up
    b. $P = spray\ pressure$
    c. Unit: PSI
    d. Nozzle + Size + P determines expected operating point

**Look up derived characteristics**

These values do not need independent sensors and can be retrieved from the nozzle guide based on selected operating point.

4. Flow Rate
    a. $Q = nozzle\ flow\ rate$
    b. Unit: GPM
    c. $Q = f(nozzle\ size,\ P)$
5. Droplet size classification
    a. Chart provides spray-quality classification such as
        i. UC - ultra coarse
        ii. XC - extremely coarse
        iii. VC - very coarse
        iv. C - coarse
        v. M - medium
        vi. F - fine
        vii. VF - very fine
    b. Classifications are typically based on standard droplet size classification systems such as ASABE S572.1 / ISO 25358
    c. Data type: categorical / ordinal
    d. Droplet size strongly influences:
        i. Settling velocity
        ii. Drift susceptibility
        iii. Coverage density
        iv. Evaporation sensitivity
    e. Key relationship: $Nozzle + P \rightarrow Droplet\ Class$
    f. Not fixed
6. Spray angle / pattern
    a. Spray angle provided from nozzle
        i. Affects spray pattern width and overlap
    b. $Ws = 2Htan(\alpha/2)$
        i. Ws = theoretical spray width
        ii. H = boom height

iii. a = nominal spray angle

7. Application rate
    a. Included on chart and useful for validation but not necessary as an independent model input since app rate can be calculated from a previously stated formula
    b. $flow\ rate\ +\ travel\ speed\ +\ spray\ width\ \rightarrow\ application\ rate$

CoverageAI Implementation

$$Selected\ Nozzle\ Family\ +\ Size$$
$$\downarrow$$
$$Measured\ Pressure$$
$$\downarrow$$
$$Manufacturer\ Lookup\ Table$$
$$\downarrow$$
$$Q,\ Droplet\ Class,\ \alpha,\ Nozzle\ Technology$$

These then join the other CovageAI inputs:

$$Wind\ Speed$$
$$Wind\ Direction$$
$$T,\ RH$$
$$Travel\ Speed$$
$$Boom\ Height$$
$$Pressure$$
$$Nozzle\ -\ Derived\ Characteristics$$

and feed:

$$CoverageAI\ Prediction\ Model$$

which predicts:

$$Coverage,\ Target\ Deposition,\ Spray\ pattern\ deviation/displacement$$

**Primary interactions**

Characteristics should be considered with:

$$Nozzle\ x\ P$$

Because pressure changes both flow and droplet behavior

$$Droplet\ Class\ x\ U$$

Because smaller droplets are more susceptible to wind transport

$$\alpha\ x\ H$$

Because spray angle and boom height determine pattern width and overlap

$$Q\ x\ v$$

Because flow rate and travel speed together determine application rate

**Intentionally excluded factors**

- Internal nozzle geometry
- Pattern type
- Fluid breakup mechanics
- CFD inside the nozzle
- Discharge coefficient
- Ligament information
- Nozzle wear

**Critical relationship to highlight**

Nozzle Relationship:

$$Nozzle\ Family\ +\ Nozzle\ Size\ +\ P\ \rightarrow\ Q\ +\ Droplet\ Class$$

Connecting resulting variables to the rest of the system:

$$Q\ +\ Droplet\ Class\ +\ \alpha\ \rightarrow\ Coverage/\ Deposition\ /\ Pattern\ deviation$$

**Conclusion**: Nozzle characteristics are configuration-based spray-system inputs. For CoverageAI, nozzle family and nozzle size will primarily serve as lookup identifiers rather than standalone physical predictors. Manufacturer application guides will be used to map the selected nozzle and measured spray pressure to operating characteristics including flow rate, droplet size classification, spray angle, and nozzle technology. These derived characteristics will then be supplied to the prediction model along with environmental and machine parameters. This approach limits the project to five documented nozzle families while preserving the physical differences most relevant to coverage, deposition, and drift.

# Spray Pressure:

| Field | Definition |
|---|---|
| Parameter | Spray Pressure |
| Category | Spray system / Operational |
| Unit | PSI |
| Role | Continuous model input and controllable operating parameter |
| Controllable? | Yes |
| Measured or Derived? | Measured |
| Expected Range | 20-90 PSI based on given nozzle ranges |
| Primary Relationships | Pressure affects nozzle flow rate, droplet-size classification, spray angle/pattern behavior, and application rate |
| Output Affected | Flow rate, droplet characteristics, application rate, coverage, target deposition, off-target deposition/drift |
| Source/ Data Needed | Pressure sensor plus nozzle family/size lookup table |

**Spray pressure**:

For a fixed nozzle:

$$P \uparrow \Rightarrow Q \uparrow$$

where:

- P = spray pressure
- Q = nozzle flow rate

TeeJet's spray pressure guidance gives the following relationship

$$\frac{Q1}{Q2} = \frac{\sqrt{P1}}{\sqrt{P2}}$$

or

$$Q2 = Q1\frac{\sqrt{P2}}{\sqrt{P1}}$$

Interpretation

$$Q \propto \sqrt{P}$$

**Pressure and Droplet Size**

Pressure changes droplet characteristics, as pressure increases, droplet size decreases:

$$P \uparrow \Rightarrow d \downarrow$$

For a given nozzle configuration where d represents characteristic droplet size. However the model should use the nozzle manufacturer data to determine droplet size given a nozzle and pressure.

**ASABE droplet classification**
ANSI/ASABE S572.3 defines droplet-spectrum categories for comparing spray nozzles and states that the classification is intended primarily to indicate off-site spray drift potential and secondarily application efficacy.

These classes include:

- VF - very fine
- F - fine
- M - medium
- C - Coarse
- VC - very coarse
- XC / EC - extremely coarse
- UC - ultra coarse

Pressure indirectly influences drift through:

$$P \rightarrow Droplet\ class \rightarrow Drift\ susceptibility$$

**Recommended model treatment:**

$$P = continuous\ pressure\ input\ (PSI)$$

$$Nozzle\ Family + Nozzle\ Size + P$$

To derive

$$Q \text{ and } Droplet\ Class$$

**Primary Interactions:**

$$P\ x\ Nozzle\ Family$$

Because pressure effects differ by nozzle design

$$P\ x\ Nozzle\ Size$$

Because different orifice sizes produce different flow and droplet behavior

$$P \rightarrow Q$$

Which affects application rate

$$P \rightarrow Droplet\ Class$$

Which affects drift and coverage and indirectly

$$P\ x\ U$$

Because pressure-driven droplet-sizes changes alter sensitivity to wind

**Conclusion**: Spray pressures is a continuous and controllable spray-system parameter that influences nozzle flow rate, droplet-size classification, spray pattern, and application rate. For a fixed nozzle, flow rate varies approximately with the square root of pressure, $Q2 = Q1\sqrt{P2/P1}$. Increasing pressure generally increases nozzle flow and tends to decrease droplet size, although the magnitude of these effects depends on the nozzle family and size. CoverageAI will therefore use measured pressure together with nozzle family and size to retrieve pressure dependent flow rate and droplet classification from manufacturer performance tables. Spray pressure should be modeled jointly with nozzle characteristics, travel speed, wind conditions, and boom height rather than as an independent predictor of coverage drift.

Sources:
https://www.teejet.com/-/media/dam/agricultural/usa/sales-material/catalog/cat51a_us.pdf
https://elibrary.asabe.org/abstract.asp?aid=51101

# Flow Rate and Application Rate:

| Field | Flow Rate | Application Rate |
|---|---|---|
| Category | Spray system | Spray System / Derived |
| Unit | GPM or L/m | GPA or L/ha |
| Role | Continuous model input / lookup-derived operating variable | Derived operating feature |
| Controllable? | Yes, indirectly through pressure/ nozzle control | Yes, indirectly through pressure/ nozzle control |
| Measured or Derived? | Measured | Derived |
| Expected Range | Nozzle and pressure dependent | Application specific |
| Primary Relationships | Depends on nozzle size/type and pressure | Depends on flow rate, travel speed, and treated width |
| Output Affected | Application rate, coverage, deposition | Coverage, target deposition, application uniformity |
| Source/ Data Needed | Flow sensor and/or manufacturer chart | Calculated from flow, speed, and width |

**Flow rate**:

Flow rate can be derived from the machine

$$q = spray\ flow\ rate$$

Pressure-flow behavior was defined earlier in the document.

Conceptually: $Nozzle + P \rightarrow q$

**Application Rate:**

ASABE published biosystems engineering reference gives:

$$AR = \frac{600q}{ws}$$

where:

- q = liquid flow rate, L/min
- AR = application rate, L/ha
- w = working width, m
- s = travel speed, km/h

**Interaction:**

$$AR \propto q$$

and

$$AR \propto \frac{1}{s}$$

Conclusion: Flow rate is a continuous spray-system parameter representing the volume of liquid delivered per unit time. It may be measured directly or obtained from the selected nozzle's manufacturer performance table at the

current spray pressure. Application rate represents the amount of spray applied per unit ground area and is treated as a derived feature rather than an independent sensor input. Using the ASABE liquid-application relationship, application rate is calculated from flow rate, travel speed, and treated width as $AR = 600q / (ws)$. Flow rate and application rate primarily influence target deposition, spray coverage, and application uniformity.

# Validating Outputs

## Coverage

| Field | Definition |
|---|---|
| Output | Spray Coverage |
| Category | Validated output / spray performance |
| Unit | % |
| Role | Model prediction target and post-spray validation metric |
| Controllable? | Not directly, influenced by spray parameters |
| Measured or Derived? | Derived from post-spray image analysis |
| Expected Range | 0-100% |
| Primary Relationships | Influenced by application rate, droplet size, nozzle characteristics, pressure, travel speed, boom height, and environmental conditions |
| Validation Method | WSP analyzed using USDA-ARS depositscan |
| Source/ Data Needed | WSP, USDA-ARS deposit scan software |

**Definition**:
The percentage of the analyzed water sensitive paper surface covered by detected spray deposits. USDA-ARS has a program called DepositSCan that evaluates spray-deposit distribution on water-sensitive paper or other deposit cards.

**Validation Method: USDA-ARS DepositScan**

Experimental validation workflow will use:

$$WSP \rightarrow Scanner \rightarrow USDA\ DepositScan$$

DepositScan can report several spray-deposit metrics including:

- Percent coverage
- Number of deposits
- deposits/cm^2
- Dv0.1
- Dv0.5
- Dv0.9
- Individual spot measurements
- Estimated deposition per unit area

$$Cobs = DepositScan\ Percentage\ Coverage$$

**Prediction versus validation**

$$\widehat{C} = predicted\ percent\ coverage$$

$$Cobs \;=\; observed\ percent\ coverage\ from\ WSP$$

**Coverage prediction error**

A signed coverage error can be calculated as

$$ec \;=\; Cobs \;-\; \widehat{C}$$

where:

- ec < 0: CoverageAI overpredicted coverage
- ec > 0: CoverageAI underpredicted coverage
- ec = 0: prediction matched measured coverage

Model performance evaluation:

$$AEc \;=\; \left|Cobs \;-\; \widehat{C}\right|$$

**Closed-loop role**

1. Prediction
    a. Current conditions are supplied to the model
        i. Wind, T, RH, travel speed, boom height, nozzle, pressure, flow / application rate, droplet characteristics
        ii. Which produces $\widehat{C}$
    b. Application and verification
        i. Spray → WSP → DepositScan → Cobs
    c. Comparison

**Relationship to model inputs**

CoverageAI will learn a relationship conceptually represented by:

$$\widehat{C} \;=\; f(U,\ \theta,\ T,\ RH,\ v,\ H,\ P,\ Q,\ AR,\ nozzle\ characteristics,\ droplet\ characteristics)$$

**Conclusion**: Spray coverage is a continuous validated output representing the percentage of an analyzed target surface contacted by spray deposits. In CoverageAI experiments, coverage will be measured using water-sensitive paper and USDA-ARS DepositScan software. DepositScan analyzes scanned WSP samples and reports percentage area coverage along with other spray-distribution statistics. The CoverageAI model will predict percent coverage $\widehat{C}$, while DepositScan percent coverage will provide the observed validation value Cobs. Prediction error can then be calculated by comparing predicted and observed coverage. Coverage is treated separately from the deposition because percent area covered does not necessarily indicate the total quantity of spray deposited on the target.

Sources:
https://www.ars.usda.gov/ARSUserFiles/50820500/Publications/Zhu_2010_DepositScan.pdf

# Target deposition

| Field | Definition |
|---|---|
| Output | Target deposition |
| Category | Validated output / spray performance |
| Unit | Estimated spray volume per unit area, typically $uL/cm^2$, if using DepositScan-derived deposition |
| Role | Model prediction target and post-spray validation metric |
| Controllable? | Not directly, influenced by spray parameters |
| Measured or Derived? | Estimated from WSP measurement |
| Expected Range | No universal fixed range |
| Primary Relationships | Influenced by application rate, droplet size, nozzle characteristics, pressure, travel speed, boom height, wind temperature, and RH |
| Validation Method | WSP scans |
| Source/ Data Needed | WSP, USDA-ARS deposit scan software |

**Definition**:
Estimated spray volume deposited on the intended target per unit area.

Formulas:

Individual droplet volume:

$$Vi = \frac{\pi d_i^3}{6}$$

Total estimated deposited volume:

$$Vtotal = \sum_{i=1}^{n} Vi$$

Deposition per unit area:

$$D = \frac{Vtotal}{A}$$

where:

- D = estimated deposition
- Vi = estimated volume of each droplet
- A = analyzed WSP area

For model validation:

$$ed = Dods - \widehat{D}$$

where Dobs is the DepositScan-estimated deposition and $\widehat{D}$ is CoverageAI predicted deposition.

Necessary assumptions:

- WSP is used as a representative target surface.
- DepositScan's droplet-size estimates are used to calculate deposited volume

- Droplets are approximated as spherical when converting diameter to volume.
- The same WSP type, scanner settings, and spray solution are used consistently
- DepositScan deposition is treated as an estimated comparative measurement, not exact deposited mass, because stain spreading affects estimated droplet diameter. USDA specifically notes that spread factor depends on spray solution properties.

**Conclusion**: Target deposition is the estimated spray volume deposited per unit target area. It will be derived from WSP measurements using USDA-ARS DepositScan. DepositScan estimates droplet diameters from stain measurements, allowing individual droplet volumes to be approximated using $V = \pi d3^3/6$. Total estimated deposited volume is divided by the analyzed WSP area to obtain deposition per unit area. This output is distinct from coverage because coverage measures surface area contacted whereas deposition estimates the quantity of spray delivered.

Sources:
https://www.ars.usda.gov/research/publications/publication/?seqNo115=337949

# Spray pattern deviation / displacement

| Field | Definition |
|---|---|
| Output | Spray pattern deviation |
| Category | Validated output / spray performance |
| Unit | mm or cm of spatial shift |
| Role | Model prediction target and post-spray validation metric |
| Controllable? | Not directly, influenced by spray parameters |
| Measured or Derived? | Estimated from WSP measurement |
| Expected Range | Treatment and distance dependent; ideally approaches 0 with increasing downwind distance. |
| Primary Relationships | Wind speed/direction, boom height, droplet size, nozzle characteristics, pressure, travel speed, temperature/ RH |
| Validation Method | WSP scans |
| Source/ Data Needed | WSP deposition values plus known distance and direction from the target/application area |

**Definition for CoverageAI**:
Spray pattern deviation describes how the observed spray deposited on the WSP differs from the expected deposition pattern under the selected nozzle configuration.

**Expected versus observed pattern:**
For each configuration, CoverageAI has an expected response:

$$Cexpected$$

After spraying DepositScan provides:

$$Cobserved$$

Coverage deviation:

$$\Delta C = Cobserved - Cexpected$$

But:

$$\Delta C < 0 \Rightarrow / wind\ displacement$$

**Spatial displacement:**
If the WSP image retains spatial information, the center of the observed deposited pattern can be calculated.

For a binary or weighted spray image, the centroid is:

$$xc = \frac{m10}{m00}, yc = \frac{m01}{m00}$$

Where mij terms are image spatial moments. This is the standard image-centroid relationship documented by OpenCV.

If the expected center of deposition is: $(xe, ye)$
And the observed center is $(x0, y0)$
Then:

$$\Delta x = xo - xe$$
$$\Delta y = yo - ye$$

The total displacement magnitude is:

$$\Delta r = \sqrt{(\Delta x)^2 + (\Delta y)^2}$$

**Wind to interpret the shift**

$$pattern\ shift + wind\ direction\ agreement + coverage\ deficit \rightarrow evidence\ of\ wind - related\ displacement$$

**Assumptions:**

- The WSP is placed at a known and repeatable expected spray location.
- WSP orientation is recorded so displacement direction can be compared with wind direction.
- The expected spray pattern for the nozzle/configuration is known or established from a baseline condition.
- WSP size and placement remain consistent between tests.
- DepositScan is used for standard coverage and droplet measurements. USDA states that DepositScan reports percent coverage and other spray-deposit characteristics from selected WSP image areas
- Low coverage alone is not classified as displacement.
- Directional pattern shift consistent with measured wind strengthens the displacement interpretation.
- If essentially no spray reaches the card, the system can identify a major deviation but cannot determine the exact distance the spray traveled.

**Conclusion**: Spray pattern deviation/displacement is a validated output describing differences between the expected spray pattern and the pattern observed on a single standardized WSP feedback card. Percent coverage from USDA-ARS DepositScan is used to identify coverage deficits, while spatial analysis of the deposited pattern can be used to estimate directional shift relative to the expected pattern location. Wind speed and direction are then used as supporting environmental information when interpreting whether the observed deviation is consistent with wind-induced displacement. A coverage deficit alone is not considered proof of displacement, and the single WSP is treated as a local feedback measurement rather than a measure of total spray drift.

Sources:
https://docs.opencv.org/3.3.0/d8/d23/classcv_1_1Moments.html